

Physical Optics: Double-Slit and Multiple-Slit Interference

PHYS 2203 — Introductory Physics III

September 14, 2026

Overview

Until now light has been treated as rays (*geometrical optics*). We now turn to *physical optics*, in which light is treated as waves that can **interfere** and **diffract**. We analyze two-slit interference using the wave model, introduce the complex plane and Euler's relation as tools for adding oscillating functions, and then generalize to N -slit interference and the diffraction grating.

1 Geometrical Optics versus Physical Optics

- * **Geometrical optics:** treat light as rays.
- * **Physical optics:** treat light as *waves* that can interfere and diffract.

Ray optics is the short-wavelength limit of the wave description: it works as long as every aperture and obstacle is large compared to λ . Once the features of the system become comparable to the wavelength, the wave nature of light can no longer be ignored.

2 Double-Slit Interference

Consider monochromatic, coherent light incident on a barrier with two narrow slits separated by a distance d . The slits sit a distance ℓ from a viewing screen, and we measure the field at a point P a height y above the central axis.

Wave fronts. A **wave front** is a surface of constant phase. For instance, the set of all points where the wave is at a crest at one instant. A small source emits spherical wave fronts (drawn as circles in the plane of the figure below). Successive fronts are one wavelength λ apart, and each front advances at the wave speed. Rays are everywhere perpendicular to the wave fronts — this is how the ray picture of the previous lectures connects to the wave picture used in physical optics.

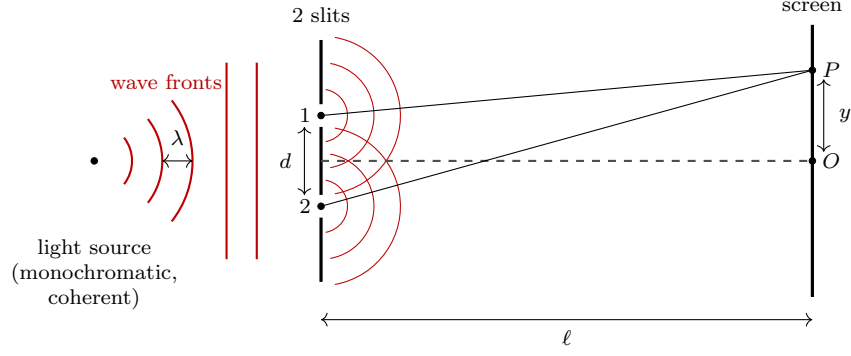


Figure 1: Double-slit interference. Far from the source the wave fronts are approximately planar. The two slits act as point-like sources of light with the same ω and k . The fields they emit overlap and interfere at the screen.

Far-field assumptions.

- * Far from the source, the wave fronts are approximately planar.
- * The slits act as point-like sources (with the same ω and k).

Let r_1 and r_2 be the distances from slit 1 and slit 2 to the point P . Each slit radiates a field

$$E_1(r_1, t) = E_0 \sin(kr_1 - \omega t), \quad r_1 = \text{distance from slit 1}, \quad (1)$$

$$E_2(r_2, t) = E_0 \sin(kr_2 - \omega t), \quad r_2 = \text{distance from slit 2}. \quad (2)$$

What do we see on the screen?

A **bright fringe** appears where E_1 and E_2 *constructively* interfere. A **dark fringe** appears where they *destructively* interfere.

Constructive interference. The two fields add in phase, $E_1(P, t) = E_2(P, t)$. This requires the phase difference to be an integer multiple of 2π :

$$(kr_2 - \omega t) - (kr_1 - \omega t) = (\text{integer}) 2\pi, \quad (3)$$

$$k(r_2 - r_1) = (\text{integer}) 2\pi. \quad (4)$$

Destructive interference. The fields cancel, $E_1(P, t) = -E_2(P, t)$. This requires the phase difference to be an odd multiple of π :

$$(kr_2 - \omega t) - (kr_1 - \omega t) = (\text{odd integer}) \pi, \quad (5)$$

$$k(r_2 - r_1) = (\text{odd integer}) \pi. \quad (6)$$

3 Path Difference: Geometry

We now use geometry to find the path difference $r_2 - r_1$.

Assumption $\ell \gg d$. When the screen is far away, $\vec{r}_1$ is nearly parallel to $\vec{r}_2$, and the two ray angles are essentially equal, $\theta_1 \approx \theta_2 \equiv \theta$. Dropping a perpendicular from slit 1 onto ray 2 forms a small right triangle whose hypotenuse is d and whose short side is the path difference:

$$|r_1 - r_2| = d \sin \theta. \quad (7)$$

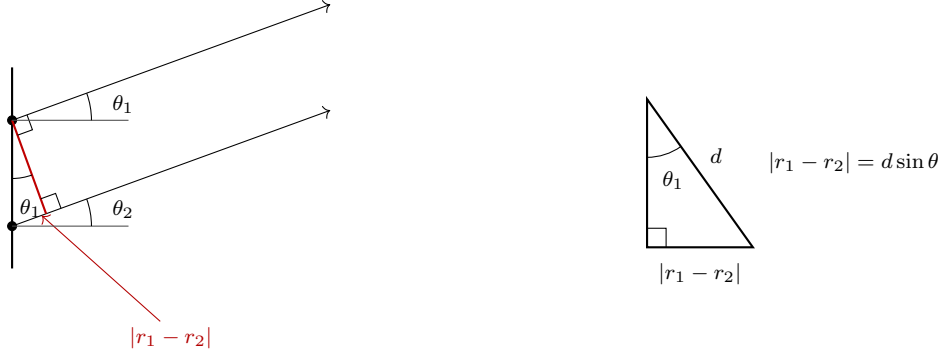


Figure 2: For $\ell \gg d$ the rays from the two slits are nearly parallel and make a common angle θ . The path difference is the short leg of a right triangle with hypotenuse d , giving $|r_1 - r_2| = d \sin \theta$.

Condition for bright fringes

Constructive interference occurs when the path difference equals an integer number of wavelengths. From $d \sin \theta = (\text{integer}) \frac{2\pi}{k}$ and $k = 2\pi/\lambda$,

$$d \sin \theta = m \lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (8)$$

For small angles, $d \theta \approx m \lambda$.

Position on the screen. At the point P , the height is $y = \ell \tan \theta \approx \ell \theta$ for small θ .

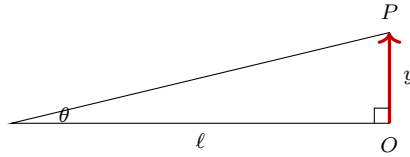


Figure 3: Geometry relating the fringe angle θ to the screen position $y = \ell \tan \theta \approx \ell \theta$.

Combining $\theta \approx m\lambda/d$ with $y \approx \ell \theta$ gives the locations of the maxima:

$$y \approx \frac{\ell \lambda}{d} m. \quad (9)$$

The fringes are equally spaced, with separation $\ell\lambda/d$ between neighboring maxima.

Condition for dark fringes

Destructive interference requires the path difference to be an odd number of half-wavelengths, $k(r_2 - r_1) = (\text{odd integer})\pi$. The same substitution gives

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (10)$$

The dark fringes fall exactly halfway between the bright ones, at $y \approx \frac{\ell\lambda}{d} \left(m + \frac{1}{2}\right)$ on the screen.

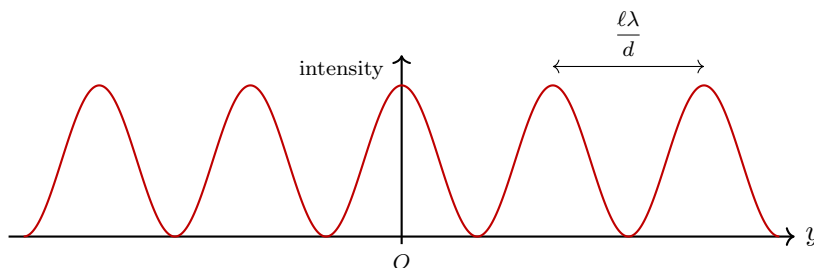


Figure 4: Two-slit intensity pattern on the screen: equally spaced bright fringes separated by $\ell\lambda/d$. Maxima occur at $y \approx (\ell\lambda/d)m$.

4 Complex Plane and Euler's Relation

The **complex plane** and **Euler's relation** are useful mathematical tools for studying oscillating functions.

Euler's relation.

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad (11)$$

where θ is a real number and $i = \sqrt{-1}$ is imaginary. The left side $e^{i\theta}$ is complex; $\cos \theta$ is its real part and $\sin \theta$ is its imaginary part.

Complex plane. Represent $e^{i\theta}$ as a vector in the plane whose horizontal axis is the real axis and whose vertical axis is the imaginary axis. The vector has unit length and makes an angle θ with the real axis, so its tip traces a circle of radius 1.

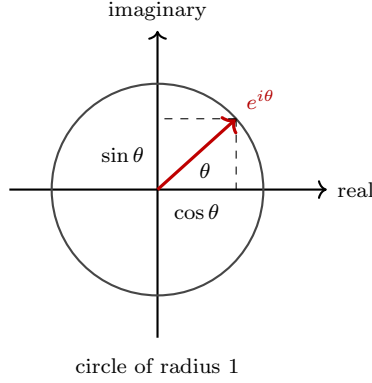


Figure 5: The number $e^{i\theta}$ as a unit vector in the complex plane. Its real part is $\cos \theta = \text{Re}\{e^{i\theta}\}$ and its imaginary part is $\sin \theta = \text{Im}\{e^{i\theta}\}$.

The real and imaginary parts are extracted as

$$\cos \theta = \text{Re}\{e^{i\theta}\} \text{ ("real part of } e^{i\theta}\text{")}, \quad \sin \theta = \text{Im}\{e^{i\theta}\} \text{ ("imaginary part of } e^{i\theta}\text{")}. \quad (12)$$

Rotating vector. If $\theta = \omega t$, then the vector rotates in the counterclockwise direction with time:

$$e^{i\omega t} = \cos \omega t + i \sin \omega t. \quad (13)$$

This gives two complementary ways to add oscillating functions:

- Add oscillating functions *mathematically* $\rightarrow$ use algebra.
- Add oscillating functions *as vectors* (phasors) $\rightarrow$ use geometry.

5 Multiple-Slit Interference

Now consider N identical slits, each separated from its neighbor by d , with the screen far away ($\ell \gg d$).

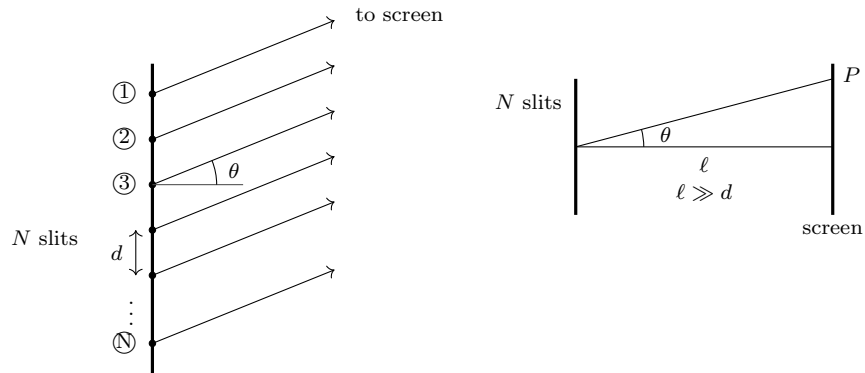


Figure 6: N -slit geometry. Neighboring slits are separated by d . The path difference between neighboring rays is $r_{n+1} - r_n = d \sin \theta$, with $n = 1, 2, 3, \dots, N - 1$.

The path difference between neighboring rays is $r_{n+1} - r_n = d \sin \theta$. Summing the field from all N slits at the point P ,

$$\begin{aligned} E(P, t) &= E_0 \left[\sin(kr_1 - \omega t) + \sin(kr_2 - \omega t) + \sin(kr_3 - \omega t) + \cdots + \sin(kr_N - \omega t) \right] \\ &= E_0 \left[\sin(kr_1 - \omega t) + \sin(kr_1 - \omega t + \delta) + \sin(kr_1 - \omega t + 2\delta) \right. \\ &\quad \left. + \cdots + \sin(kr_1 - \omega t + (N-1)\delta) \right], \end{aligned} \quad (14)$$

where

$$\delta = \left(\frac{2\pi}{\lambda} \right) d \sin \theta = k d \sin \theta \quad (15)$$

is the phase difference between neighboring rays.

Evaluating the sum with the complex plane

Write the sum as the imaginary part of a geometric series of phasors:

$$E(P, t) = E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta} + e^{i2\delta} + \cdots + e^{i(N-1)\delta}) \right\}. \quad (16)$$

Check, $N = 2$. For two slits,

$$E(P, t) = E_0 \left[\sin(kr_1 - \omega t) + \sin(kr_1 - \omega t + \delta) \right] = E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta}) \right\}. \quad (17)$$

Adding the two phasors E_1 and E_2 (each of amplitude E_0 , with relative phase δ) head-to-tail forms an isosceles triangle whose base is the resultant.

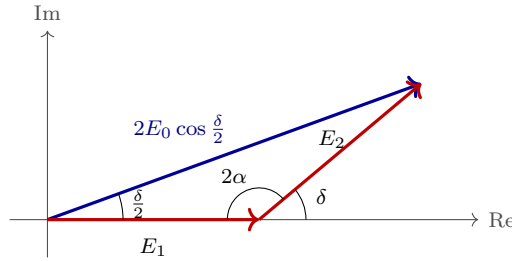


Figure 7: Phasor addition for $N = 2$, with E_1 drawn along the real axis. Each phasor has amplitude E_0 , and E_2 is turned from E_1 by the phase difference δ . The interior angle at the joint is 2α and the exterior turn is δ , so $2\alpha + \delta = \pi$ and $\alpha = \frac{\pi}{2} - \frac{\delta}{2}$. The resultant is the base of the isosceles triangle: it has length $2E_0 \cos(\delta/2)$ and makes angle $\delta/2$ with E_1 , so it equals $2E_0 \cos(\frac{\delta}{2}) e^{i\delta/2}$.

Resultant amplitude from the triangle. The interior angle at the joint plus the exterior turn δ make a straight line, so $2\alpha + \delta = \pi$, hence

$$\alpha = \frac{\pi}{2} - \frac{\delta}{2}. \quad (18)$$

Drop a perpendicular from the joint onto the resultant. Because the triangle is isosceles ($E_1 = E_2 = E_0$), this line bisects both the apex angle 2α and the base, splitting the triangle into two

congruent right triangles, each with hypotenuse E_0 and apex half-angle α . The half-base is the side opposite α , so

$$\frac{1}{2} |E| = E_0 \sin \alpha = E_0 \sin \left(\frac{\pi}{2} - \frac{\delta}{2} \right) = E_0 \cos \frac{\delta}{2}, \quad (19)$$

using $\sin(\frac{\pi}{2} - x) = \cos x$. The resultant amplitude is therefore $|E| = 2E_0 \cos(\delta/2)$.

The field. From the triangle, the phasor sum $1 + e^{i\delta}$ is a vector of magnitude $2 \cos(\delta/2)$ (the resultant length, with the common E_0 factored out) pointing at angle $\delta/2$ (the bisector). Since a vector of magnitude M at angle φ is written $M e^{i\varphi}$,

$$1 + e^{i\delta} = \underbrace{2 \cos \frac{\delta}{2}}_{\text{magnitude}} \underbrace{e^{i\delta/2}}_{\text{rotated by } \delta/2}. \quad (20)$$

Substituting this and evaluating the imaginary part,

$$\begin{aligned} E(P, t) &= E_0 \operatorname{Im} \left\{ e^{i(kr_1 - \omega t)} (1 + e^{i\delta}) \right\} \\ &= \operatorname{Im} \left\{ 2E_0 \cos \frac{\delta}{2} e^{i(kr_1 - \omega t + \delta/2)} \right\} \\ &= 2E_0 \cos \frac{\delta}{2} \sin \left(kr_1 - \omega t + \frac{\delta}{2} \right). \end{aligned} \quad (21)$$

The intensity is the time average of E^2 , and since $\langle \sin^2 \rangle = \frac{1}{2}$ is a constant,

$$\text{Intensity} \sim \langle E^2(P, t) \rangle \propto \left(2E_0 \cos \frac{\delta}{2} \right)^2, \quad (22)$$

which is maximized when $\frac{\delta}{2} = m\pi$, i.e.

$$\frac{1}{2} \frac{2\pi}{\lambda} d \sin \theta = m\pi \quad \implies \quad d \sin \theta = m\lambda \quad \checkmark \quad (23)$$

in agreement with the two-slit result found earlier.